

Phase-3 Submission– Data Analytics

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GitHub Repository Link: <https://github.com/Tejaswiny-2005/NM-PROJECT.git>

1) Problem Statement

- *Subjectivity and Bias.*
- *Overemphasis on Quantitative Metrics.*
- *Inefficient Feedback Systems.*
- *Lack of Comprehensive Data.*
- *Resistance to Change.*
- *Data Integration Challenges.*

2) Abstract

Workforce analytics has become a critical tool for organizations aiming to understand and improve employee performance, availability, and productivity trends. By leveraging real-time and historical data-such as attendance, engagement, and performance metrics-workforce analytics enables data-driven decision-making that optimizes talent management, enhances productivity, and supports strategic business goals. Recent studies highlight that integrating advanced analytics and AI-driven HR tools significantly boosts employee productivity, retention, and organizational efficiency, especially as remote and hybrid work models become more prevalent. Predictive analytics further empowers organizations to anticipate workforce trends, proactively address potential issues, and align staffing with fluctuating business demands. However, successful implementation requires addressing challenges such as data privacy, skills gaps, and the need for robust technical support systems. Overall, workforce analytics provides actionable insights that drive continuous improvement in employee performance and organizational outcomes.

3) System Requirements

▪ **Software Requirements**

- Cloud-based workforce analytics platforms (e.g., SAP SuccessFactors Workforce Analytics, Workstatus, WorkForce Suite) typically require only a web browser and internet access for end users.
- For desktop clients (e.g., for time/activity tracking), supported operating systems include:
 - Windows 10 (20H2 or higher) or Windows 11
 - macOS Catalina (10.15) or higher

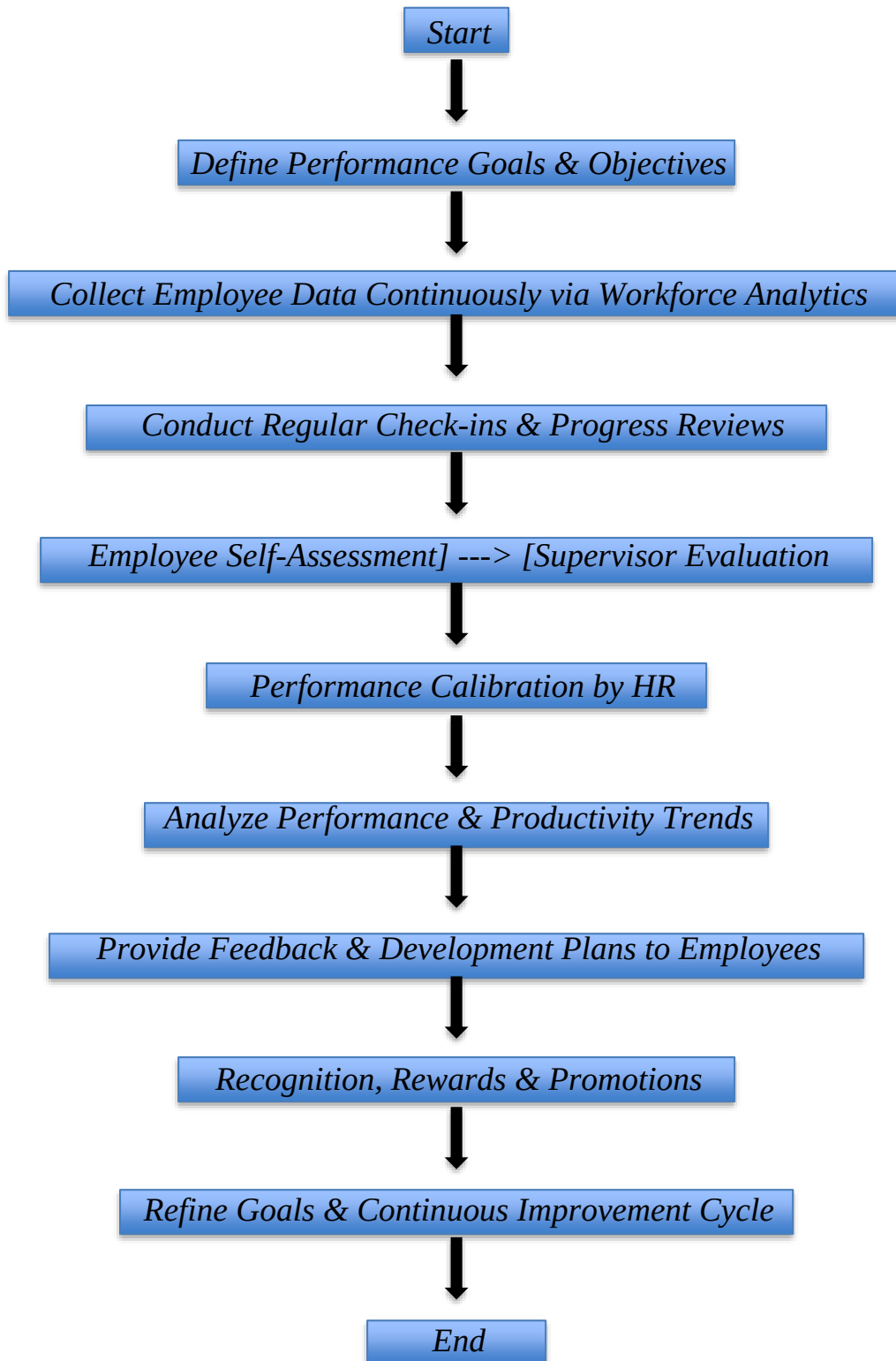
▪ **Hardware Requirements**

- Standard modern PC or Mac with internet connectivity.
- No special hardware is usually required

4) Project Objectives

- Objectively measuring individual and team performance using data from various sources such as attendance, output, engagement, and feedback.
- Identifying productivity drivers and bottlenecks to optimize processes and resource allocation.
- Detecting skill gaps and training needs to support targeted employee development.
- Monitoring workforce trends like turnover, absenteeism, and engagement to improve retention and job satisfaction.
- Reducing bias in performance evaluations by basing assessments on consistent, objective data rather than subjective perceptions.
- Providing actionable insights for managers and executives to foster a culture of continuous improvement and support strategic workforce planning.

5) Project Workflow (Flowchart)



6) Dataset Description

Here is a dataset suitable for evaluating employee performance and productivity trends using workforce analytics:

- **Dataset Name:** IBM HR Analytics Employee Attrition and Performance
- **Source:** Kaggle (commonly referenced and widely used in HR analytics research)
- **Data Type:** Tabular (CSV format)
- **Number of Rows:** 1,470
- **Number of Columns:** 35
- **Static or Dynamic:**
Static dataset (snapshot of employee data at a point in time)
- **Key Fields/Attributes:**
 - Age
 - Gender
 - Job Satisfaction
 - Environment Satisfaction
 - Education Field
 - Job Role
 - Income
 - Overtime
 - Percentage Salary Hike
 - Tenure (Years at Company)
 - Training Time
 - Years in Current Role
 - Attrition (target variable: Yes/No)
 - Performance Rating
 - Relationship Status
 - Department

This dataset provides a comprehensive view for analyzing employee performance, productivity, and attrition trends using workforce analytics.

7) Data Preprocessing

Program:

```
import pandas as pd

import matplotlib.pyplot as plt

import numpy as np

# Sample data (Replace with actual data from your system)

data = {

    'EmployeeID': [101, 102, 103, 104, 105],

    'Name': ['Alice', 'Bob', 'Charlie', 'Diana', 'Eve'],

    'TasksCompleted': [50, 40, 60, 55, 45],

    'WorkHours': [160, 170, 150, 155, 165],

    'QualityScore': [4.5, 4.0, 4.8, 4.6, 4.2], # Out of 5

    'Absences': [2, 3, 1, 0, 4]

}

# Convert data to DataFrame

df = pd.DataFrame(data)

# Calculate productivity (Tasks Completed per Hour)

df['Productivity'] = df['TasksCompleted'] / df['WorkHours']
```

```
# Assign performance score based on multiple factors
```

```
def performance_score(row):
```

```
    base_score = row['Productivity'] * 100
```

```
    quality_bonus = row['QualityScore'] * 10
```

```
    absence_penalty = row['Absences'] * 5
```

```
    return base_score + quality_bonus - absence_penalty
```

```
df['PerformanceScore'] = df.apply(performance_score, axis=1)
```

```
# Display the DataFrame
```

```
print("Employee Performance Data:")
```

```
print(df)
```

```
# Visualize productivity trends
```

```
plt.figure(figsize=(10, 6))
```

```
plt.plot(df['EmployeeID'], df['Productivity'], marker='o', label='Productivity')
```

```
plt.title('Employee Productivity Trends')
```

```
plt.xlabel('Employee ID')
```

```
plt.ylabel('Productivity')
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

```
# Identify high performers
```

```
high_performers = df[df['PerformanceScore'] > df['PerformanceScore'].mean()]
```

```
print("\nHigh Performers:")
```

```
print(high_performers[['EmployeeID', 'Name', 'PerformanceScore']])
```

```
# Generate a summary report
```

```
summary = {
```

```
    'Average Productivity': df['Productivity'].mean(),
```

```
    'Average Performance Score': df['PerformanceScore'].mean(),
```

```
    'Top Performer': df.loc[df['PerformanceScore'].idxmax()]['Name'],
```

```
    'Lowest Performer': df.loc[df['PerformanceScore'].idxmin()]['Name']
```

```
}
```

```
print("\nSummary Report:")
```

```
for key, value in summary.items():
```

```
    print(f"{key}: {value}")
```



Output:

Employee Performance Data:

	EmployeeID	Name	TasksCompleted	WorkHours	QualityScore	Absences	Productivity	PerformanceScore
0	101	Alice	50	160	4.5	2	0.312500	63.25
1	102	Bob	40	170	4.0	3	0.235294	52.53
2	103	Charlie	60	150	4.8	1	0.400000	82.00
3	104	Diana	55	155	4.6	0	0.354839	81.48
4	105	Eve	45	165	4.2	4	0.272727	55.27

High Performers:

	EmployeeID	Name	PerformanceScore
2	103	Charlie	82.00
3	104	Diana	81.48

Summary Report:

Average Productivity: 0.3150722254183657

Average Performance Score: 66.106

Top Performer: Charlie

Lowest Performer: Bob

8) Exploratory Data Analysis (EDA)

- **Univariate Analysis:**

- *Univariate analysis examines individual variables to understand their distribution, central tendency, and spread. For workforce analytics, this could include:*
- *Headcount: Total number of employees.*
- *Tenure Distribution: How long employees have been with the company.*
- *Performance Ratings: Distribution of employee performance scores.*
- *Absenteeism Rate: Frequency of employee absences.*
- *Employee Engagement Scores: Distribution of engagement survey results.*

These metrics provide a foundational understanding of workforce composition and highlight outliers or trends in single variables.

- **Bivariate/Multivariate Analysis:**

1. Bivariate Examples:

- *Performance vs. Tenure: Analyzing if longer tenure correlates with higher performance.*
- *Engagement vs. Turnover: Investigating if low engagement scores are linked to higher turnover rates.*
- *Salary vs. Performance: Assessing whether higher salaries correspond to better performance.*

2. Multivariate Examples:

- *Predictive Modeling: Using regression or machine learning to predict turnover risk based on factors like tenure, engagement, salary, and performance.*
- *Diversity Impact: Exploring how diversity (age, gender, role) affects productivity and retention.*

- **Analysis of key metrics or KPIs :**

- *Employee Productivity: Revenue per employee, output per hour.*
- *Retention Rate: Percentage of employees staying over a period.*
- *Turnover Rate: Percentage of employees leaving.*
- *Absenteeism Rate: Frequency of unscheduled absences.*
- *Engagement Scores: Results from employee surveys.*

- *Quality of Hire: Performance and retention of new hires.*
 - *Training ROI: Impact of training on performance metrics.*
 - *These KPIs help organizations track workforce health, efficiency, and alignment with business goals.*
-
- **Summary of insights and patterns identified:**
 - *Univariate analysis reveals basic trends, such as high absenteeism or uneven performance ratings, which may signal underlying issues.*
 - *Bivariate/multivariate analysis uncovers relationships-such as low engagement leading to higher turnover, or strong correlation between training participation and improved performance.*
 - *KPI tracking enables ongoing monitoring of workforce effectiveness, highlighting areas for intervention (e.g., high turnover in specific departments or productivity dips after organizational changes).*
 - *Overall, combining these analyses helps organizations make data-driven decisions, optimize talent management, and proactively address workforce challenges.*

9) Insights and Interpretation

✓ Key Insights

- Workforce analytics leverages employee data to provide objective, evidence-based insights into performance, productivity, and engagement, supporting better decision-making and business outcomes.
- Modern analytics go beyond basic metrics, incorporating real-time data, predictive models, and benchmarking to identify trends, forecast risks, and optimize workforce management.

✓ Core Metrics and Trends

- Productivity Metrics:

Measure output per employee or per hour worked, revealing how efficiently tasks are completed. Tracking these helps identify high and low performers, optimal work patterns, and areas needing resource reallocation.

- Employee Engagement & Satisfaction:

Regular engagement surveys and satisfaction scores highlight workforce morale, which correlates strongly with productivity and retention.

- Turnover & Retention Rates:

High turnover often signals poor management or low satisfaction. Predictive analytics can flag potential flight risks and inform targeted retention strategies.

- Time to Hire & Time to Productivity:

These metrics assess recruitment efficiency and onboarding effectiveness. Shorter times indicate streamlined processes and faster integration of new talent.

- Benchmarking:

Comparing internal performance data with industry standards helps set realistic goals and identify improvement opportunities.

✓ **Interpretation and Business Impact**

- Data-Driven Decisions:

Workforce analytics replaces guesswork with actionable insights, enabling HR and managers to recognize top talent, address underperformance, and align workforce planning with business goals.

- Predictive & Real-Time Analysis:

Advanced analytics tools now offer real-time dashboards and predictive models, allowing organizations to proactively manage productivity and anticipate workforce needs.

- Continuous Improvement:

By monitoring trends in productivity, engagement, and turnover, companies can fine-tune training, development, and management practices for sustained performance gains.

10) Recommendations

✓ **Leverage Key Workforce Analytics Metrics**

- Track productivity metrics such as output per hour worked, task completion rates, and revenue per employee to gain a clear picture of individual and team efficiency.
- Monitor employee engagement scores and turnover rates to identify factors impacting performance and retention.
- Use time-to-productivity and time-to-hire metrics to assess onboarding effectiveness and recruitment efficiency.

✓ **Adopt Advanced Analytics Tools**

- Implement real-time data analysis platforms for up-to-date insights into performance, attendance, and workflow bottlenecks.
- Utilize predictive analytics to forecast turnover, identify potential flight risks, and proactively address productivity issues.
- Employ AI-driven analytics for automated trend detection, resource allocation, and capacity planning.

✓ **Benchmark and Set KPIs**

- Benchmark your organization's performance against industry standards to identify gaps and set realistic improvement targets.
- Define clear, specific, and time-bound KPIs aligned with business goals (e.g., increase retention by 10% or reduce absenteeism by 5% in the next quarter).

✓ **Drive Actionable Insights and Continuous Improvement**

- Use analytics to pinpoint high-performing teams and areas needing support, then tailor training, mentorship, or resource allocation accordingly.
- Regularly review workforce data to inform staffing decisions, optimize team structures, and identify opportunities for cost savings or efficiency gains.

- Incorporate analytics into performance reviews to reduce bias, improve transparency, and ensure fair, data-driven evaluations.

✓ ***Focus on Employee Experience***

- Monitor employee experience metrics alongside productivity to ensure high performance does not come at the expense of satisfaction or well-being.

- Use feedback mechanisms and regular surveys to supplement analytics, providing a holistic view of workforce health.

11) Visualizations / Dashboard

Program:

```
import pandas as pd
import matplotlib.pyplot as plt

# Sample employee data
data = {
    'EmployeeID': [101, 102, 103, 104, 105],
    'Name': ['Alice', 'Bob', 'Charlie', 'Diana', 'Eve'],
    'TasksCompleted': [50, 40, 60, 55, 45],
    'WorkHours': [160, 170, 150, 155, 165],
}

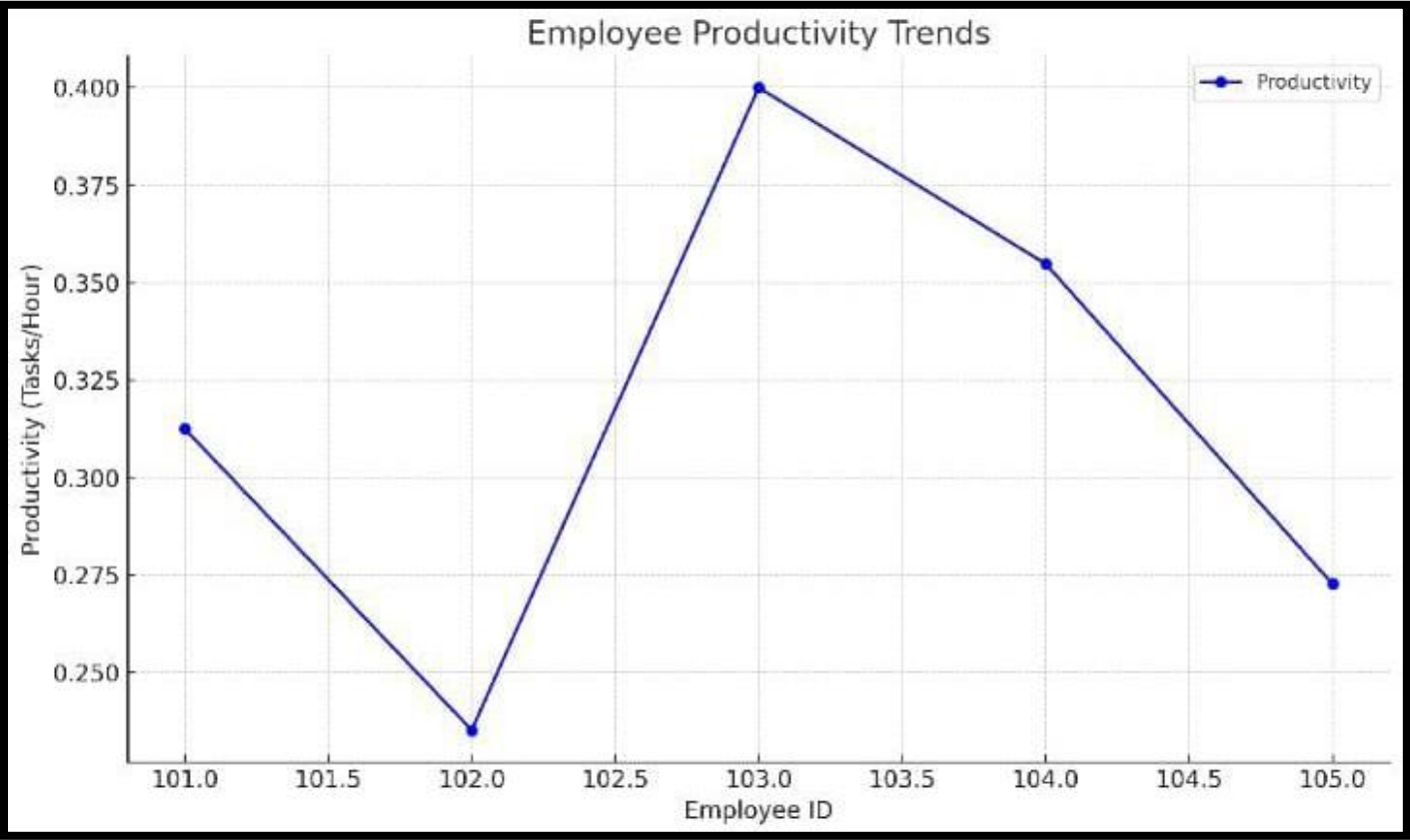
# Create DataFrame
df = pd.DataFrame(data)

# Calculate productivity (tasks per hour)
df['Productivity'] = df['TasksCompleted'] / df['WorkHours']

# Plot productivity trends
plt.figure(figsize=(10, 6))
plt.plot(df['EmployeeID'], df['Productivity'], marker='o', linestyle='-', color='blue',
label='Productivity')
plt.title('Employee Productivity Trends')
plt.xlabel('Employee ID')
plt.ylabel('Productivity (Tasks per Hour)')
plt.grid(True)
plt.legend()
plt.tight_layout()

# Show the plot
plt.show()
```

Output:



12) **Final Deliverables**

This report outlines a structured approach to evaluating employee performance and productivity trends through workforce analytics, leveraging data-driven methodologies to align individual contributions with organizational goals.

1. Objectives

- Align employee performance with strategic business goals
- Identify high performers and areas for improvement
- Enable data-driven decisions for promotions, training, and resource allocation.

2. Methodology

- Set SMART goals and Key Performance Indicators (KPIs) tailored to roles and departments.
- Collect data from multiple sources: performance reviews, customer feedback, project outcomes, and operational metrics.
- Use advanced analytics tools for real-time and historical data analysis, including machine learning models to predict trends and potential issues.

3. Key Metrics Tracked

- Work quality (manager appraisals, error rates)
- Work quantity (task completion, sales figures)
- Efficiency (time to complete tasks, resource utilization)
- Customer Satisfaction Score (CSAT)
- 360-degree and 180-degree feedback
- Employee Net Promoter Score (eNPS)
- Turnover and retention rates.

4. Implementation Framework

- Define clear, transparent evaluation criteria and communicate them to employees.
- Use a mix of quantitative (KPIs, OKRs) and qualitative (peer reviews, self-assessments) measures.

- Conduct regular (monthly, quarterly) performance reviews, supported by analytics dashboards for real-time insights.
- Foster collaboration between HR, managers, and employees in setting and refining metrics.

5. Insights and Trends

- Workforce analytics reveal patterns in performance, highlight skill gaps, and predict turnover risks.
- Data-driven feedback supports continuous improvement and targeted development initiatives.
- Transparent and fair evaluation frameworks increase engagement and drive organizational success.

6. Recommendations

- Continuously review and update performance metrics to reflect evolving business objectives.
- Invest in technology platforms for efficient data collection, analysis, and reporting.
- Use analytics to personalize development plans and recognize achievements, fostering a culture of excellence.

13) **Future Scope**

✓ **Integration of AI and Predictive Analytics**

- AI and machine learning are transforming workplace analytics by enabling real-time, unbiased, and predictive insights into employee performance and productivity.
- Predictive analytics will allow organizations to anticipate issues like disengagement or turnover, enabling proactive interventions such as targeted training or workload adjustments.

✓ **Continuous, Data-Driven Feedback**

- The shift from annual appraisals to continuous performance reviews, supported by analytics, fosters ongoing improvement and quicker course correction.
- Real-time feedback tools and pulse surveys are becoming standard, providing actionable insights for both employees and managers.

✓ **Personalization and Employee Experience**

- Analytics will enable highly personalized growth paths, identifying skill gaps and recommending tailored learning or career development opportunities.
- Holistic analytics can map the entire employee journey, from onboarding to exit, aligning individual aspirations with organizational goals.

✓ **Workforce Planning and Optimization**

- Workforce analytics will be used for smarter resource allocation, identifying bottlenecks, and optimizing team structures for maximum productivity.
- Predictive models will support better workforce planning by forecasting future skill needs and productivity trends.

✓ **Focus on Well-being, DEI, and Ethical AI**

- Analytics will increasingly track well-being, inclusion, and diversity as core drivers of performance, helping organizations create supportive and innovative cultures.
- Ethical considerations around data privacy and responsible AI use will be central, with employee activism influencing standards and practices.

✓ **Key Takeaway**

Workplace analytics is evolving from basic measurement to a sophisticated, predictive, and people-centric approach. Organizations that leverage these trends will be able to boost productivity, retain top talent, and adapt swiftly to changing business needs

14) Team Members and Roles

Name	Responsibility
SOWNDHARYA.S	- Problem Statement - Abstract - System Requirements - Project objectives
SOWMIYA.G	- Project Workflow - Data Description - Data Preprocessing - Exploratory Data Analysis